AI-Powered Meeting Intelligence and Task Management System

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1. Problem Statement

Organizations conduct numerous internal meetings daily, resulting in large volumes of discussion that are difficult to document manually. Employees often spend significant time creating meeting minutes, tracking action items, assigning responsibilities, and monitoring deadlines. Manual notetaking can lead to missed information, inaccurate documentation, forgotten tasks, and inefficient project management.

Additionally, tracking deadlines and integrating meeting outcomes into calendar systems requires extra effort, reducing productivity and increasing the likelihood of missed commitments.

Therefore, there is a need for an intelligent system that can automatically convert meeting audio into structured meeting notes, identify tasks and deadlines, assign responsibilities, and synchronize important events with calendar applications while maintaining secure user access.

2. Introduction

The **AI-Powered Meeting Intelligence and Task Management System** is a web-based application designed to automate the process of meeting documentation and task management.

The system allows authorized users to upload recorded meeting audio files through a secure web portal. Once uploaded, the audio undergoes speech-to-text conversion using the **Faster-Whisper** model. The generated transcript is then processed using the **BART Transformer Model** from Hugging Face to generate concise meeting summaries.

Advanced Natural Language Processing techniques are further applied to extract important information such as:

- Assigned tasks
- Responsible employees
- Deadlines
- Task priorities

The extracted deadlines are automatically integrated into calendar systems through generated ICS calendar files and synchronization with Google Calendar. Users can view task deadlines, update completion status, and manage meeting histories through an interactive dashboard.

The platform also incorporates **JWT-based Authentication and Authorization**, ensuring that only authorized users can access and manage meeting data.

The solution significantly reduces manual effort, improves documentation accuracy, enhances productivity, and ensures effective follow-up of meeting action items.

3. Objectives

Primary Objectives

- Automate meeting transcription.
- Generate concise meeting summaries.
- Extract actionable tasks from discussions.
- Identify task owners and deadlines.
- Integrate tasks with calendar systems.
- Provide secure user authentication.
- Maintain meeting history for future reference.

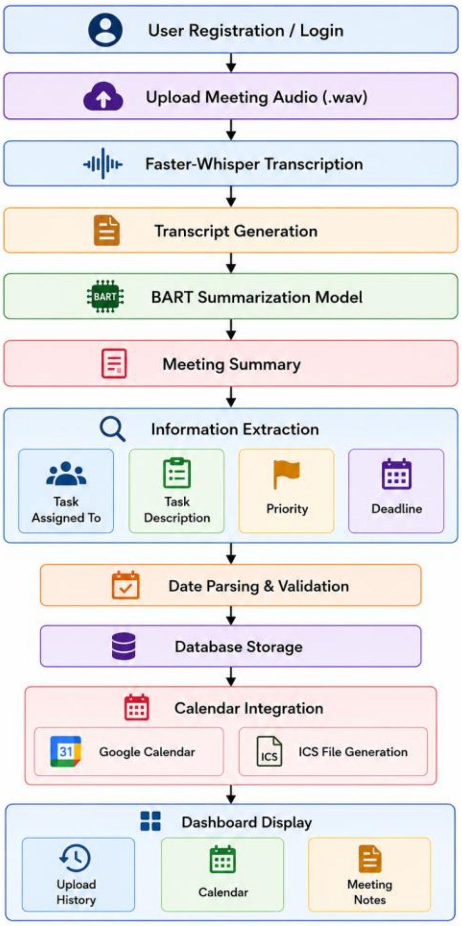
4. Proposed Solution

The proposed system provides an end-to-end automated meeting management workflow.

Users upload recorded meeting audio files through the web application. The backend processes the audio using AI and NLP models to generate transcripts, summaries, deadlines, priorities, and assigned tasks. Extracted information is displayed through a dashboard and synchronized with calendar applications.

The system maintains a history of all processed meetings and allows users to monitor task completion status through an integrated calendar interface.

5. System Workflow



6. Web Application (Frontend)

The frontend is developed using **React.js** and provides an intuitive user interface.

Modules

1. Login Page

- User authentication
- JWT token validation
- Secure access control

2. Registration Page

- New user registration
- User credential storage
- Account creation

3. Dashboard

Contains navigation cards for:

- Upload Meeting
- Calendar

- Meeting History
- Logout

4. Upload Meeting Page

- Upload WAV audio files
- Processing status display
- generated results display

5. Calendar Module

- View deadlines
- Update task status
- Track completion
- Synchronize events

6. History Module

- Display previously uploaded meetings
- Delete summaries
- View summary based on dates
- View summaries
- Edit generated meeting summaries before saving or sharing.
- Share meeting summaries via Gmail directly from history module to any email recipient.

7. Algorithms and Models Used

1. Faster-Whisper

Purpose

Speech-to-text transcription.

Working

- Accepts WAV audio files.
- Converts spoken content into text.
- Optimized implementation of Faster-Whisper.
- Faster inference with lower memory consumption.

Benefits

- High transcription accuracy.
- Fast processing speed.
- Supports long meeting recordings.

2. BART Transformer Model

Purpose

Meeting summarization.

Working

- Receives transcript text.
- Produces concise meeting summaries.

Benefits

- Context-aware summarization.
- Reduces lengthy transcripts.
- Captures important discussion points.

3. Regular Expressions (Regex)

Purpose

Task extraction.

Working

Pattern matching identifies:

- Task owner
- Assigned activity
- Priority keywords

Example:

```
John will complete API integration by Friday.
```

Extracts:

```
Assigned                              To:                              John
Task: API Integration
```

4. DateParser

Purpose

Deadline extraction.

Working

Converts natural language dates such as:

```
Tomorrow
Next                                                                 Monday
15 July 2026
```

into standardized datetime formats.

5. JWT Authentication Algorithm

Purpose

Secure authentication.

Working

- User logs in.
- Server validates credentials.
- JWT token generated.
- Token attached to future requests.
- Backend verifies token before granting access.

8. Database Design

User Collection

```
User ID
Name
Email
Password
```

Meeting Collection

```
Meeting ID
User ID
Audio File Path
Transcript
Summary
Upload Date
```

Task Collection

```
TaskID
MeetingID
AssignedTo
Task Description
Priority
Deadline
Status
```

9. Performance Optimization Techniques

Faster-Whisper Optimization

- Uses optimized inference engine.
- Reduced memory consumption.
- Faster processing compared to original Whisper.

Efficient API Design

- RESTful architecture.
- Modular backend implementation.

JWT Session Management

- Stateless authentication.
- Reduced database lookups.

Database Optimization

- Indexed user and meeting records.
- Faster retrieval of meeting history.

Frontend Optimization

- React component reuse.
- Efficient state management.
- Lazy loading of pages.

10. Advantages

Business Advantages

- Eliminates manual note-taking.
- Improves meeting productivity.
- Reduces human errors.
- Enhances accountability.

Technical Advantages

- Automated transcription.
- Accurate summarization.
- Deadline extraction.
- Calendar synchronization.
- Secure JWT authentication.
- Meeting history management.

User Advantages

- Easy-to-use interface.
- Quick access to meeting information.
- Better task tracking.
- Improved deadline management.
- Edit generated summary before finalizing.
- Share meeting summaries instantly through gmail.

11. Limitations

- Accuracy depends on audio quality.
- Multiple speakers may reduce transcription accuracy.
- Complex sentence structures may affect task extraction.
- Calendar synchronization requires internet access.
- Very long meetings may increase processing time.
- Regex-based extraction may miss unusual task formats.

12. Future Enhancements

AI Improvements

- Speaker diarization (identify who spoke).
- Multi-language transcription.
- Emotion and sentiment analysis.
- Advanced task extraction using Large Language Models.

Productivity Enhancements

- Real-time meeting recording.
- Live transcription during meetings.
- Automatic email notifications.
- Slack and Microsoft Teams integration.

Calendar Enhancements

- Two-way Google Calendar synchronization.
- Outlook Calendar integration.
- Automated reminder notifications.

Analytics Enhancements

- Meeting participation analytics.
- Team productivity reports.
- Task completion statistics.
- Dashboard visualizations.

Security Enhancements

- Role-Based Access Control (RBAC).
- Multi-factor authentication (MFA).

13. Conclusion

The **AI-Powered Meeting Intelligence and Task Management System** provides a comprehensive solution for automating meeting documentation, task extraction, and deadline management. By leveraging **Faster-Whisper**, **BART Transformers**, **Regex**, **DateParser**, and **JWT Authentication**, the system transforms meeting recordings into actionable insights with minimal human intervention.

The integration of calendar management, task tracking, secure authentication, and historical record maintenance makes the platform highly suitable for organizations seeking to improve operational efficiency, collaboration, and accountability. The proposed solution significantly reduces manual effort while ensuring that critical meeting outcomes are accurately captured, tracked, and completed on time.